

Hemachandrika Rage

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PROFESSIONAL EXPERIENCE

BI Data Analyst Dec 2021 – Dec 2022

Tata Consultancy Services, Bengaluru, Karnataka, India

- Developed executive-level Power BI dashboards to improve audit visibility, expedite financial reporting, and enable real-time tracking of key performance indicators (KPIs) for the airline sector.
- Initiated SQL queries for data extraction and preprocessing that were performance-optimized, cutting down on report creation time by 20% and speeding up operational and financial decision-making.
- Implemented ETL workflows to enhance data pipelines and accuracy, saving approximately 20 hours of manual labor per week and ensuring seamless integration of airline-specific data sources.
- Conducted compliance reporting and advanced analytics in accordance with airline industry standards, improving compliance accuracy by 30% and reducing late submission trends, which increased operational efficiency.

Data Analyst Intern Nov 2020 – Nov 2021

Myzas Technologies PVT LTD

- Produced thorough summary reports for internal audit teams at clients that manufacture batteries, offering vital information that improved compliance and expedited audit procedures.
- Leveraged Power BI, Python scripts, and SQL Server to combine data from various sources to develop key performance indicators (KPIs) that decreased manual data processing time by 20% and increased reporting accuracy by 25%.
developed Tableau interactive dashboards that enhanced data-driven decision-making, increasing stakeholder use of BI tools by 20%.
- Broadened the precision and effectiveness of financial analysis and reporting through analyzing process statistics from the purchase-to-payment cycle and creating sample criteria for compliance.
- Collaborated with production managers to do cohort analysis of asset data, improving overall financial reporting and accuracy while reducing valuation discrepancies for a manufacturing client by 3%.

SKILLS

- Programming Languages** : Python, SQL, R, MySQL, Jupyter Notebook, VS Code, GitHub
- Visualization Tools** : Matplotlib, Seaborn, Excel, Pivot Tables, VBA, Power BI, Tableau,
- Cloud Platform** : Snowflake, AWS, Hadoop, Hive
- Data Analysis & Modeling** : EDA, Statistical Analysis, Data Cleaning, Data Preprocessing, Pandas, NumPy
- Machine Learning** : Logistic Regression, Unsupervised Learning, Supervised Learning, Predictive Modelling,
- Deep Learning Frameworks** : Keras, TensorFlow, Hypothesis Testing, Regression, BeautifulSoup,
- Natural Language Processing** : NLTK, VADER (Valence Aware Dictionary & sEntiment Reasoner), Latent Dirichlet Allocation.

PROJECTS

Sentiment Analysis- Customer Feedback Analysis for British Airways [Link](#)

Python, Web Scrapping, NLTK, LDA Feb 2024 – April 2024

- Analyzed customer reviews from airlinequality.com, extracting consumer sentiment trends specifically for British Airways.
- Developed and executed Python scripts for web scraping, data preprocessing, and sentiment analysis, improving data retrieval efficiency by 20%.
- Established detailed visualizations using Python libraries (Matplotlib and Seaborn), which facilitated a 25% faster decision-making process for marketing strategies based on consumer insights.
- Enhanced strategic communication efforts by effectively translating complex data insights into actionable recommendations, leading to a 15% improvement in campaign target alignment based on natural language processing findings.

Neural Network-based Handwritten Digit Recognition [Link](#)

Python, Keras, TensorFlow, CNN Nov 2023 – Jan 2024

- Tackled the MNIST handwritten digit classification task with dataset having 60,000 images.
- In the beginning, used a multilayer perceptron (MLP) which is also a neural network, and on the test dataset, which helped to achieve an impressive accuracy of over 96%.
- Switched to a CNN architecture with dropout regularization to improve model performance even further and investigate convolutional neural networks (CNNs), which produced even higher accuracy.
- By utilizing methods including data preparation, hyperparameter adjustment, and model assessment, the CNN model has improved to surpass 96.58% test accuracy.

EDUCATION

Saint Peter’s University, Jersey City, New Jersey May 2023 – Dec 2024

Master of Science in Data Science – CGPA 3.85/4

- Gaining expertise in data analysis, machine learning, big data analytics, and predictive modeling. Skilled in Python programming, data mining, and visualization, with hands-on experience in applying distributed computing technologies like Hadoop and Spark.